

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-af1031dfbc4b471fa.jsonl`  
- SHA-256: `78a45a0fd58dfe5720a6831c4fe3dcbc235bcad87426a7d6126a46146f936028`  
- Source modified: `2026-05-31T04:23:13+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:22:06.960Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"The Qwen Research License prohibits commercial use, which casts doubt on whether the InternVL2.5-4B weights can legally be used commercially despite the MIT repo license."

****Source:**** <https://github.com/OpenGVLab/InternVL/issues/900> (primary)

****Supporting quote:**** "Since the Qwen RESEARCH LICENSE prohibits commercial use, I would like to confirm whether the InternVL2_5-4B model is eligible for commercial use under these circumstances."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:22:10.434Z)

I'll verify this claim about the Qwen Research License and InternVL2.5-4B commercial use eligibility. Let me start by examining the primary source and gathering contradicting evidence.

3. assistant (2026-05-31T04:22:13.663Z)

I'll verify this claim about the Qwen Research License and InternVL2.5-4B. The claim has two parts: (1) the Qwen Research License prohibits commercial use, and (2) this casts doubt on InternVL2.5-4B's commercial usability despite the MIT repo license. Let me investigate both.

4. user (2026-05-31T04:22:18.254Z)

GitHub Issue #900: InternVL2.5-4B License Question

****Original Question:****

The user notes a discrepancy in licensing information. The HuggingFace page states Qwen2.5-3B-Instruct uses "Apache License 2.0," but the user indicates the actual license is the Qwen Research License, which prohibits commercial use. They ask: "I would like to confirm whether the InternVL2_5-4B model is eligible for commercial use under these circumstances."

****Maintainer Response:****

No response from maintainers is visible in the provided content. The issue shows as "Closed" but the page does not display any replies addressing the user's licensing concern about commercial eligibility.

The core issue remains unresolved in this excerpt?whether the commercial use restriction from Qwen's research license affects InternVL2.5-4B's usability despite its own MIT license designation.

5. user (2026-05-31T04:22:20.543Z)

Web search results for query: "InternVL2.5-4B license Qwen2.5 base model commercial use restriction"

Links: [{"title": "About the license of internVL2.5-4B model · Issue #900 · OpenGVLab/InternVL", "url": "https://github.com/OpenGVLab/InternVL/issues/900"}, {"title": "LICENSE · Qwen/Qwen2.5-VL-72B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE"}, {"title": "LICENSE · Qwen/Qwen2.5-72B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-72B-Instruct/blob/main/LICENSE"}, {"title": "LICENSE · Qwen/Qwen2.5-7B at main", "url": "https://huggingface.co/Qwen/Qwen2.5-7B/blob/main/LICENSE"}, {"title": "Qwen Promises AI Agent Capabilities From It Free-to-Use Qwen2.5-VL Vision Language Models - Hackster.io", "url": "https://www.hackster.io/news/qwen-promises-ai-agent-capabilities-from-it-free-to-use-qwen2-5-vl-vision-language-models-db51344a23b4"}, {"title": "LICENSE · Qwen/Qwen2.5-0.5B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct/blob/main/LICENSE"}, {"title": "LICENSE · Qwen/Qwen2.5-VL-3B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE"}, {"title": "LICENSE · Qwen/Qwen2.5-3B at main", "url": "https://huggingface.co/Qwen/Qwen2.5-3B/blob/main/LICENSE"}, {"title": "LICENSE · Qwen/Qwen2.5-14B-Instruct at main", "url": "https://huggingface.co/Qwen/Qwen2.5-14B-Instruct/blob/main/LICENSE"}]

Based on the search results, I found relevant information about the licensing issue:

The InternVL2_5-4B model on Hugging Face claims to be released under the MIT License and uses the pre-trained Qwen2.5-3B-Instruct as a component, which is stated as being licensed under the Apache License 2.0. However, the actual license for Qwen2.5-3B-Instruct is not Apache 2.0, but rather the Qwen RESEARCH LICENSE.

Since the Qwen RESEARCH LICENSE prohibits commercial use, there is a question about whether the InternVL2_5-4B model is eligible for commercial use under these circumstances.

This appears to be a licensing discrepancy that has been raised as a concern on the project's GitHub repository, as there is a conflict between the stated license information on Hugging Face and the actual restrictions imposed by the Qwen base model's research license.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:22:23.973Z)

The claim is well-grounded so far. The key facts I need to verify precisely: (1) Does the Qwen2.5-3B-Instruct (the base model for InternVL2.5-4B) actually use the Qwen Research License rather than Apache-2.0? (2) Does that license actually prohibit commercial use? Let me check the actual Qwen2.5-3B license file and the InternVL2.5-4B model card directly.

7. user (2026-05-31T04:22:29.931Z)

InternVL2.5-4B License and Components

Model License

The model card states the license is **MIT**.

Base/Language Model

InternVL2.5-4B uses **Qwen2.5-3B-Instruct** as its language component.

From the model family table:

Model Name	Vision Part	Language Part
InternVL2.5-4B	InternViT-300M-448px-V2_5	Qwen2.5-3B-Instruct

Qwen2.5-3B-Instruct License

Yes, the model card explicitly mentions Qwen2.5-3B-Instruct and attributes an **Apache License 2.0** to it:

> This project is released under the MIT License. This project uses the pre-trained Qwen2.5-3B-Instruct as a component, which is licensed under the Apache License 2.0.

This means while the overall InternVL2.5-4B project is MIT-licensed, the Qwen2.5-3B-Instruct component must be used in compliance with Apache License 2.0.

8. user (2026-05-31T04:22:30.285Z)

Qwen Research License Agreement

License Type

This is the **Qwen RESEARCH LICENSE AGREEMENT**, a restrictive research-focused license from Alibaba Cloud.

Commercial Use Restrictions

Yes, it **prohibits commercial use** without a separate license. Here are the exact clauses:

Primary Restriction (Section 2.a):

> "You are granted a non-exclusive, worldwide, non-transferable and royalty-free limited license under Alibaba Cloud's intellectual property or other rights owned by us embodied in the Materials to use, reproduce, distribute, copy, create derivative works of, and make modifications to the Materials **FOR NON-COMMERCIAL PURPOSES ONLY**."

Commercial Licensing Requirement (Section 2.b):

> "If you are commercially using the Materials, you shall request a license from us."

Key Restrictions on Derivative Works (Section 4.b):

> "If you use the Materials or any outputs or results therefrom to create, train, fine-tune, or improve an AI model that is distributed or made available, you shall prominently display 'Built with Qwen' or 'Improved using Qwen' in the related product documentation."

Important Notes

- **No Monthly Active User (MAU) thresholds** are mentioned in this license
- The license is **non-transferable**
- You must **request a separate commercial license** from Alibaba Cloud for any commercial use
- The license specifically defines "Non-Commercial" as "for research or evaluation purposes only" (Section 1)

This is a more restrictive license than permissive open-source licenses like Apache 2.0 or MIT.

9. assistant (2026-05-31T04:22:36.568Z)

This is a strong confirmation. Let me lay out what I've established:

1. **InternVL2.5-4B uses Qwen2.5-3B-Instruct** as its language component (confirmed on the official model card).
2. **The InternVL model card claims Qwen2.5-3B-Instruct is "Apache License 2.0"** ? but this is incorrect.
3. **Qwen2.5-3B-Instruct actually ships under the Qwen RESEARCH LICENSE** (confirmed by the actual LICENSE file on HuggingFace), which states "FOR NON-COMMERCIAL PURPOSES ONLY" and "If you are commercially using the Materials, you shall request a license from us."

This is a notable detail: the 3B size of Qwen2.5 specifically is the Qwen Research License, unlike the 0.5B/1.5B/7B/14B/32B/72B sizes. Let me verify the 3B size is genuinely the exception (an important nuance, since many Qwen2.5 sizes ARE Apache-2.0), and confirm the license file isn't outdated by checking the current state.

10. user (2026-05-31T04:22:41.181Z)

Based on the web page content provided, the license listed in the metadata/tags for the Qwen/Qwen2.5-3B-Instruct model is **qwen-research**.

This is clearly shown in the model card header where it states:

> License: qwen-research

11. user (2026-05-31T04:22:42.156Z)

Web search results for query: "Qwen2.5 3B model Qwen Research License not Apache 2.0 which sizes"

Links: [{"title": "Qwen - Wikipedia", "url": "https://en.wikipedia.org/wiki/Qwen"}, {"title": "GitHub - QwenLM/Qwen3: Qwen3 is the large language model series developed by Qwen team, Alibaba Cloud. · GitHub", "url": "https://github.com/QwenLM/qwen3"}, {"title": "Qwen2.5: A Party of Foundation Models! | Qwen", "url": "https://qwenlm.github.io/blog/qwen2.5/"}, {"title": "Qwen2.5-Coder Series: Powerful, Diverse, Practical - Alibaba Cloud Community", "url": "https://www.alibabacloud.com/blog/qwen2-5-coder-series-powerful-diverse-practical_601765"}, {"title": "Alibaba Releases Qwen 2.5 Models, Raising the Bar for Open Weight LLMs", "url": "https://www.deeplearning.ai/the-batch/alibaba-releases-qwen-2-5-models-raising-the-bar-for-open-weight-llms"}, {"title": "qwen2.5", "url": "https://ollama.com/library/qwen2.5"}, {"title": "Qwen Studio", "url": "https://qwen.ai/blog?id=qwen2.5-llm"}, {"title": "LICENSE · Qwen/Qwen2.5-7B at main", "url": "https://huggingface.co/Qwen/Qwen2.5-7B/blob/main/LICENSE"}]

Based on the search results, here's what I found about Qwen2.5 3B model licensing:

All open-source models in Qwen2.5, except for the 3B and 72B variants, are licensed under Apache 2.0. This means the 3B and 72B models are free, but their license requires special arrangements for commercial use.

For additional context on specialized variants: Qwen2.5-Coder 0.5B / 1.5B / 7B / 14B / 32B are licensed under Apache 2.0, while 3B is under Qwen-Research license.

So to directly answer your query: The **3B and 72B** are the sizes that do NOT use Apache 2.0 licensing in the Qwen2.5 family. The 3B model specifically uses the Qwen Research License (or Qwen License), while all other sizes (0.5B, 1.5B, 7B, 14B, 32B) use Apache 2.0.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-05-31T04:22:55.902Z)

I now have comprehensive verification. Let me assess the claim against the checklist.

****Findings:****

1. ****The base model is correctly identified.**** InternVL2.5-4B uses Qwen2.5-3B-Instruct as its language component (official model card confirms).
2. ****The Qwen Research License does prohibit commercial use.**** The actual LICENSE file on HuggingFace for Qwen2.5-3B-Instruct states "FOR NON-COMMERCIAL PURPOSES ONLY" (Section 2.a) and "If you are commercially using the Materials, you shall request a license from us" (Section 2.b). The license metadata field is `qwen-research`.
3. ****The 3B size specifically is the exception.**** Of the Qwen2.5 family, only the 3B and 72B sizes use the Qwen Research License; the others (0.5B, 1.5B, 7B, 14B, 32B) are Apache-2.0. So InternVL2.5-4B (built on 3B) lands precisely on a restricted base.
4. ****The InternVL model card's own attribution is wrong**** ? it claims Qwen2.5-3B-Instruct is "Apache License 2.0," which contradicts Qwen's actual license file. This is exactly the discrepancy the issue #900 author flagged.

****Checklist evaluation:****

- ***Supported by quote*** Yes. The quote accurately states the Qwen Research License prohibits commercial use and raises the commercial-eligibility doubt for InternVL2.5-4B. Verified against primary sources (the actual license file).
- ***Contradicting evidence*** None found. Multiple independent sources confirm the 3B/72B exception and the non-commercial restriction.
- ***Source quality*** The claim is backed by the primary license file itself, not just the forum issue. The forum issue is corroborative, but the substantive facts are independently verified against authoritative HuggingFace license files.
- ***Outdated*** No. The license metadata field currently reads `qwen-research` (checked the live model card).
- ***Marketing/speculation*** No. This is a documented licensing fact.

****One nuance to note for the source-quality dimension:**** The claim's framing ("casts doubt") is appropriately hedged. There IS genuine legal uncertainty here ? InternVL applies MIT to its own contributions, but a derivative work cannot grant rights to upstream weights it doesn't own. The non-commercial restriction on the Qwen base would propagate to the merged model regardless of InternVL's MIT label. The claim correctly captures this as "casts doubt" rather than asserting a definitive prohibition. The issue being closed without a maintainer answer (no resolution) further supports that the doubt is real and unresolved.

The claim is accurate, current, and backed by primary sources (the actual Qwen license file, not merely the forum post). I cannot refute it.

13. user (2026-05-31T04:23:06.833Z)

Structured output provided successfully

14. assistant (2026-05-31T04:23:13.475Z)

Verification complete. The claim is ****confirmed (not refuted)**** with high confidence.

Key evidence, verified against primary sources rather than just the cited forum issue:

- ****Qwen2.5-3B-Instruct**** (InternVL2.5-4B's confirmed language component) ships under the ****Qwen Research License****, whose Section 2.a grants rights "FOR NON-COMMERCIAL PURPOSES ONLY." The live model card's license field reads `qwen-research`.
- The ****3B size is specifically the exception**** in the Qwen2.5 family ? only 3B and 72B use the Research License; the rest are Apache-2.0. InternVL2.5-4B is built precisely on the restricted 3B.
- InternVL's own model card ****mis-attributes**** Qwen2.5-3B-Instruct as "Apache License 2.0," which is exactly the discrepancy issue #900 raised ? and the issue was closed with ****no maintainer resolution****, so the doubt is genuinely unresolved.
- The claim's "casts doubt" framing is appropriately hedged (not overclaiming a definitive prohibition), and an MIT label on a derivative cannot override the non-commercial terms of upstream weights.

Practical note for muneem's license-safety requirement: InternVL2.5-4B (and the Qwen2-based 2B variant) should be flagged commercially unsafe; the InternLM2.5-based InternVL2.5 variants (8B/26B) are the license-safe alternatives in that family.